

VENUGOPAL INDRAJITH HEMASHREE

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EDUCATION

Vellore Institute of Technology, Chennai

Aug 2023 – Present

B.Tech, Computer Science Engineering — Cyber Physical Systems | CGPA: 7.81

Maharishi International Residential School, Kanchipuram

Class 12 (CBSE): 79% | Class 10 (CBSE): 89%

PATENT

IoT-Enabled Smart Walker with Sensor-Fusion Fall Risk Prediction and Adaptive Mobility Assistance

Co-Inventor | Status: **Filing in Progress** (Inventive disclosure filed with IIPRD, Khurana & Khurana) | Faculty Guide: Dr. Subitha D

- Co-invented an Arduino Uno-based smart walker combining dual MPU6050 IMUs, dual HX711 load cells, and dual HC-SR04 ultrasonic sensors for real-time bilateral fall detection, weight-asymmetry monitoring, and obstacle avoidance, running fully standalone with no cloud dependency at an estimated hardware cost under USD 30.
- Designed a sensor-fusion Fall Risk Prediction engine and a Dynamic Stability Reserve metric that combine tilt, weight imbalance, and obstacle proximity to flag deteriorating stability before a fall occurs, alongside near-fall detection and a Gait Symmetry Index for rehabilitation tracking.

PROJECTS

AI-Based Dental Malocclusion Detection and Teeth Alignment Analysis

Jul 2025 – Sep 2025

Python, YOLOv9, Grad-CAM, Gradio, OpenCV, Deep Learning, Jupyter Notebook, GitHub

- Built a YOLOv9-based detection pipeline for automated dental malocclusion and teeth-alignment analysis from X-ray images, achieving **~97% detection accuracy across a dataset of 13,000+ images**.
- Applied Grad-CAM to visualize the regions driving each prediction, supporting clinical interpretability and orthodontic decision support.
- Deployed an interactive Gradio interface for real-time image upload, prediction, and result visualization.

Reinforcement Learning-Driven Autonomous Rover for Hazard Detection and Environmental Exploration

2026

Python · Raspberry Pi 5 · BreezySLAM · YOLOv8n / YOLOv9t · A* · Reinforcement Learning · Flask

- Engineered an autonomous ground rover that performs systematic area sweeps for disaster-response reconnaissance, running LIDAR-based SLAM (BreezySLAM) on a self-upgraded Raspberry Pi 5 to build a real-time map of the environment overlaid with a risk heatmap of detected hazard zones
- Modified the YOLOv8n architecture directly — adding a multi-task risk-score prediction head and reweighting the loss function to prioritize human detection — for real-time victim detection, running alongside a separate camera-based YOLOv9t model and an MQ-2 gas sensor for on-board fire/smoke/gas hazard detection.
- Designed a modified A* search over the live SLAM map to compute the safest and shortest evacuation path out of the hazard zone for detected humans, paired with generated voice commands to guide them to safety in real time.
- Implemented reinforcement-learning-based navigation so the rover adapts its sweep path around newly detected hazards in real time, streaming live position, sensor readings, and detections to a Flask telemetry dashboard for human oversight.
- Fused HC-SR04 ultrasonic sensors, Pi Camera, and an MPU9250 IMU across both autonomous and manual operation modes, keeping navigation robust when individual sensors are degraded.

Intrusion Detection and Threat Modeling

Nov 2025 – Jan 2026

Python, IDS, Threat Modeling, Risk Analysis, Cybersecurity, GitHub

- Designed an intrusion detection system for a simulated online banking platform, using behavioral analysis and risk-based scoring across login, access, and transaction events to flag anomalies.
- Built threat classification logic with automated response triggers, backed by detailed security logging for audit trails and post-incident analysis.

Jarvis — Local Voice-Controlled AI Assistant

Personal Project

Python, SpeechRecognition, Local System Commands

- Built a locally-run Python voice assistant that recognizes spoken commands to open applications and execute system controls (including shutdown/restart), with no cloud dependency.
- Added battery/system-status reporting and stock-price notifications, exploring fully offline voice-driven control end to end.

INTERNSHIP

Research Intern — VIT Chennai, under Prof. Pavithra

May 2025 – Jul 2025 (1 month)

Human Pose Estimation using Heatmaps (MMPose Validation) | MMPose, OpenCV, Kinovea, Matplotlib, Python

- Implemented human joint detection using heatmap-based pose estimation models with MMPose, visualizing and analyzing joint-confidence heatmaps for keypoint localization.
- Validated pose-estimation output against ground truth using Kinovea motion-analysis software to sanity-check keypoint accuracy under faculty supervision.

TECHNICAL SKILLS

Languages: Java, Python

Web Development: HTML, CSS, JavaScript

ML / Computer Vision: YOLOv8/v9, MMPose, OpenCV, Grad-CAM, TensorFlow/PyTorch, Heatmap-based Keypoint Detection

Cybersecurity: Intrusion Detection, Threat Modeling, Risk Analysis

Embedded / IoT: Arduino, Sensor Fusion

Databases & Tools: MySQL, Git, VS Code, Jupyter Notebook, Google Colab, Gradio

Spoken Languages: Tamil (Native), English (Fluent), Hindi (Intermediate), Japanese (Basic)

ACHIEVEMENTS & LEADERSHIP

- **IBM Generative AI Certification** — “Gen AI Using IBM Watsonx,” IBM Career Education Program, via IBM Developer Skills Network (Jun 2025). | [View certificate](#)
- **NUS Introduction to Industry 4.0 Programme, National University of Singapore** (Jun 2025) — completed via the School of Continuing and Lifelong Education (SCALE) & College of Design and Engineering; team member on a final project on digital transformation using the Business Model Canvas. | [View certificate](#)
- **SuperAI Conference 2025, Marina Bay Sands, Singapore** (Jun 2025) — attended Asia's largest AI conference, featuring 7,000+ innovators and 1,000+ AI companies.
- Team Leader, CodeYGen Club — led a student team through coding and design-based projects, coordinating development and presentations.
- Design Lead, Fitness Club — designed promotional creatives, posters, and digital content to support fitness events and member engagement.